MetropoLive

Group 17 - Sunsets

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https://github.com/CMPT-276-SUMMER-2025/final-project-17-sunsets

Overview:

The application will be able to find events nearby along with relevant information such as transit to the event and perhaps a calendar.

The idea for this application came from people needing a way to meet new people and go outside more. With the lack of third places nowadays, interacting with people and making connections is harder than ever. People are lonelier than ever, making them depressed. This application assists in resolving those problems. The application will also save time for those busy people who can't afford to spend time searching for events themselves. People foreign to the area will benefit from being able to easily access local events.

Potential users:

- University students looking to socialize, but have little time to search and plan due to busy schedules
- Tourists looking for events, but are unfamiliar with local transit
- Volunteers looking for opportunities

Persona for each application group: (Updated Personas)

Name: Liam
Education: Second year computer science student that transferred from Douglas College to pursue a Bachelors of Science degree at Simon Fraser University.
Age: 22
Situation & Characteristics: Liam lives in a shared apartment in Burnaby so that he is never far from campus. It's currently late June, and he's nearing the end of his midterm exams. Liam has a demanding course schedule in addition to a part-time job; however, he wants to avoid spending another summer that's entirely working and studying, and is interested in seeing what the tri-cities and Vancouver have to offer. Liam is a highly analytical and relatively introverted person. He is most comfortable in quiet, structured environments where there is a shared activity like a board game or even a hobbyist workshop that brings like-minded people together. Although not the most outgoing, once Liam has established a new branch in his daily routine, he'll stick to this without fail.
Challenges & Goals:

- Liam would like to meet like minded people and possibly make new friends, but finds it difficult to initiate conversations in large social settings like at parties or crowded bars.
- Classes and work leave Liam with very little mental (or physical) energy left at the end of the week, but still wants the ability to attend events within proximity to school and home.
- Liam is worried that most social events require upfront costs, which would conflict with his limited student budget. He's interested in mostly free events with the occasional paid venue.

Name: Sofia
Education: Recently graduated with a degree in Marketing from the University of Washington.
Age: 22
Situation: Sofia is on a week long solo trip from Seattle, and wanted to explore the greater Metro Vancouver area. She is staying at an Airbnb in Port Moody, and decided to rent an Evo hatchback for any longer day trips. Sofia plans to rely on public transit or walking to explore specific neighborhoods like downtown Maple Ridge or Vancouver, as this would limit extra charges on her rental. She loves documenting her travels on Instagram and TikTok.
Characteristics: Sofia is an outgoing and energetic individual who values crowd-sourced reviews over official tourism brochures, and loves to discover lesser-known “hidden gems” on her travels. She tends to be drawn to aesthetically appealing cafes, public art, and garden arrangements, but isn’t afraid to ask locals for their recommendations of what to do and where to visit.
Challenges & Goals:

- Since metro Vancouver and the tri-cities aren’t home to Sofia, she’s unfamiliar with both the geography and the TransLink system. Because of this, having quick access to simple and traffic-aware directions is a must.
- Sofia is eager to discover unique, photo-worthy local events, restaurants, and other small locations in between that reflect the culture of BC’s West Coast.
- Her week-long stay will go by quickly, and so Sofia is concerned about wasting precious time on overpriced or inauthentic “tourist traps”.
- Wherever possible, she would like to meet other travelers or friendly locals at social gatherings, and possibly make lasting friendships from these.

Name: Charlie
Education: Grade 11 student at Thomas Haney Secondary School in Maple Ridge.
Age: 16
Situation: It's the beginning of summer break. Charlie is proactively looking to build up his experience and volunteer hours before starting the university application process next year. He does not have a driver's license and relies entirely on his bicycle or the TransLink bus system to get around Maple Ridge and nearby communities.
Characteristics: Responsible, organized, and ambitious. He is a bit shy in new situations but thrives when he has a clear task or role to perform. He is motivated by the idea of making a tangible impact and wants to plan his summer schedule effectively to maximize his time.
Challenges & Goals:

- Finding legitimate, short-term, or single-day volunteer opportunities is difficult, as many organizations prefer long-term commitments.
- Find and participate in at least three different volunteer events over the summer to gain a variety of experiences.
- Coordinating transportation via public transit to unfamiliar event locations requires significant advance planning.
- Successfully manages his time and transportation to attend all committed events punctually and reliably.

List of chosen APIs (4 in total):

Primary:

Google Maps Platform Routes API: The official API for the familiar Google Maps interface. Like MapBox Directions, this API is responsible for calculating optimal routes between two or more locations for several modes of transportation, while also taking into account current road conditions.

Eventbrite API: The official API for Eventbrite. Eventbrite allows users to find events based on location, price, and topic. Events such as concerts, festivals, meetups, workshops, etc.

(Optional) Add to calendar button: Allows connection to calendar apps to keep track of signed up events. Instead of using another API, this feature can be implemented by generating a universal calendar file (extension ".ics") directly in the user's browser with JavaScript. A library that may be useful for this button could be "ics.js".

Secondary:

MapBox Directions API: The official API for MapBox, a provider for custom online maps. Calculates ideal routes for multiple modes of transportation, accounting for real-time traffic conditions.

Meetup API: The official API of Meetup. Meetup provides users with information on local crowd-sourced events.

Description of features for each API & User stories:

Primary:

Google Maps Platform Routes API:
- Shows optimal route between two locations
  - *"I want to see the fastest route to the event so I can get there on time for my volunteer work."*
- Shows turn-by-turn instructions
  - *"I'm not familiar with the local transit, so I want to know which buses I need to take and what stops to get off at."*
- Swap between transit profiles
  - *"I don't own a car, so I want to be able to see directions for either walking or cycling."*

Eventbrite API:

- Can search for nearby events
  - *"I want to discover new events near me so I can meet new people."*
- Can search by category
  - *"I like listening to live music, so I want to search for concerts."*
- Shows information about events such as price
  - *"I want to be able to see details about events so I can see what matches with my budget and schedule."*

Secondary:

MapBox Directions API:

- Displays optimized routes between two or more locations
  - *"I want to see the fastest routes between a handful of events that I want to attend."*
- Shows turn-by-turn instructions
  - *"The transit options near me are unfamiliar, so I want to know which buses or walking routes I need to take and what stops to make along the way."*
- Quickly switch between different modes of transit
  - *"I don't always have easy access to a car, so I want to be able to see directions for either walking, cycling, or occasionally driving."*

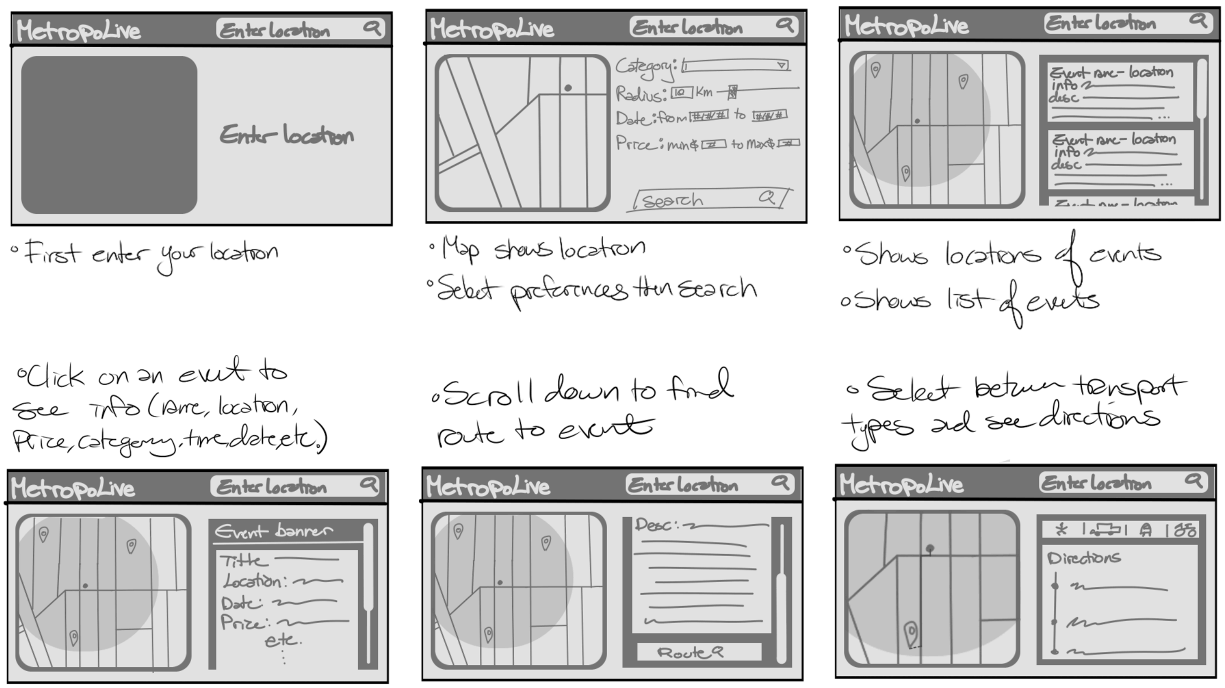
Meetup API:

- Search for events by keywords and filters
  - *"I want to be able to search for specific events that fit my interests."*
- Shows photos associated with events
  - *"I want to see photos of events so I can get a sense of what to expect before I go."*
- Provides description of events
  - *"I want a detailed description of events so I can choose one I can afford."*

Add to calendar button:

- Can connect to any major calendar app like Outlook or Apple
  - *"I need to manage all of my events effectively, so incorporating them into one app would be best."*
- Can make edits in the calendar app
  - *"I want to be able to modify or erase information quickly given that a problem occurs or the timings change."*
- Provides easy access between app and calendar
  - *"I would like for there to be a way to access the app with ease through the calendar so that I do not have to manually switch between them."*

Low-fidelity storyboard of the application interface/features: (Cleaned up storyboard)



Chosen front-end technology stack:

- React JS
- CSS modules
- HTML

Explanation why we chose this stack and how our project will use this stack: (Cleared up tech stack issue)

- Since React JS involves the process of using reusable components, our app will be using components like search bars to enter our location, event cards to show the titles, locations or images, maps and a navigation bar to navigate within the page. Further since React uses a virtual DOM we would not have to reload the whole web page as the rendering will be done only to the parts we have changed.
- The components we build will be constructed using HTML and CSS properties where the functionality will depend on JSX which is a combination of HTML and Javascript code, and the styling will be done using CSS modules. For instance building the navigation bar will depend on JSX and the overall look will depend on how we use the CSS modules within the CSS file. Moreover the designs for each component will be done separately rather than globally so the stylings from one component will not affect other components.
- Since React utilizes javascript we can use a built-in function from javascript called fetch to make requests to the APIs. The process for this would be: the user writing their search in the search bar, then react would use fetch to request for the event from an appropriate API, which would then send the event data back to

us from the API, next react will update the page with the new event data we received and display it to the user. For instance we can use the fetch function to pull in the data of events from Eventbrite and display them to our users.